CS23532 – COMPUTER NETWORKS LABORATORY

PRACTICAL – 4

AIM

To set up and configure a Local Area Network (LAN) using a network switch and Ethernet cables, assign IP addresses to the connected computers, verify connectivity using the ping command, and share a folder between computers in the LAN.

THEORY

A **Local Area Network (LAN)** is a computer network that connects devices within a limited geographical area such as a computer laboratory, office, school, or home. It allows connected devices to communicate and share resources such as files, printers, applications, and internet connections.

A **network switch** is the main connecting device used in a LAN. It connects multiple computers and forwards data between them. Each computer connected to the switch can communicate with other computers in the same network.

REQUIREMENTS

- 4 Computers
- 1 Network Switch (8/16/24 port)
- 4 Ethernet cables
- Windows operating system
- Network interface cards (Ethernet ports)

PROCEDURE

Step 1: Plan the Network

Design a suitable LAN topology based on the number of computers and available equipment. A **star topology** can be used, where all computers are connected to a central network switch.

Step 2: Connect the Devices

Take four computers, one network switch, and four Ethernet cables.

Connect one end of each Ethernet cable to the Ethernet port of the respective computer and connect the other end to separate ports on the network switch.

Step 3: Assign IP Addresses

Configure a static IPv4 address for each computer.

Go to:

Control Panel → Network and Internet → Network Connections → Ethernet → Properties → Internet Protocol Version 4 (TCP/IPv4) → Properties

Select **Use the following IP address** and enter the required IP address and subnet mask.

Use the following configuration:

Device	IP Address	Subnet Mask
PC1	10.1.1.1	255.0.0.0
PC2	10.1.1.2	255.0.0.0
PC3	10.1.1.3	255.0.0.0
PC4	10.1.1.4	255.0.0.0
Switch	10.1.1.5	255.0.0.0

The IP address of every device must be unique within the LAN.

Step 4: Configure the Network Switch

1. Connect a computer to the network switch using an Ethernet cable.
2. Open a web browser.
3. Enter the management IP address of the switch.
4. Log in using the appropriate username and password.
5. Configure the basic network settings.
6. Assign the switch management IP address as **10.1.1.5** with subnet mask **255.0.0.0**, if supported by the switch.

Step 5: Check Network Connectivity

Open **Command Prompt** on each computer.

Use the `ping` command to check connectivity between the computers.

For example, from PC1, enter:

```
ping 10.1.1.2
ping 10.1.1.3
ping 10.1.1.4
```

If the computers are properly connected and configured, replies will be received from the destination computers.

Step 6: Share a Folder

1. Select a folder on any one of the computers.
2. Right-click the folder and select **Properties**.
3. Open the **Sharing** tab.
4. Select **Advanced Sharing** or the appropriate sharing option.

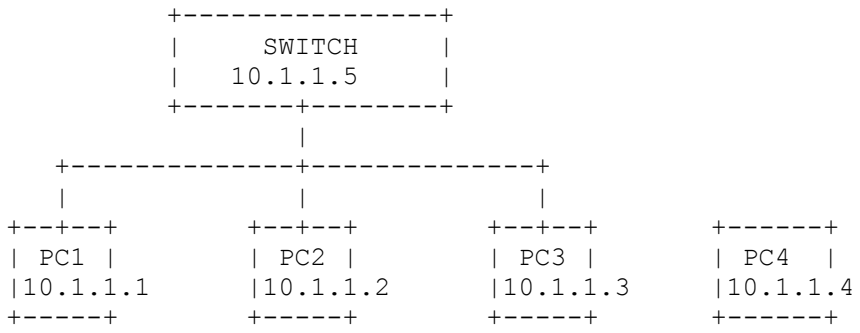
5. Enable folder sharing.
6. Configure the required permissions.
7. Allow users on the same LAN to access the shared folder.

Step 7: Access the Shared Folder

From another computer connected to the same LAN, open **File Explorer** and access the shared folder using the network path.

The shared folder should be accessible from other computers if network discovery, file sharing, permissions, and connectivity are correctly configured.

LAN CONFIGURATION DIAGRAM



IP CONFIGURATION

Device	IP Address	Subnet Mask	Connection
PC1	10.1.1.1	255.0.0.0	Ethernet
PC2	10.1.1.2	255.0.0.0	Ethernet
PC3	10.1.1.3	255.0.0.0	Ethernet
PC4	10.1.1.4	255.0.0.0	Ethernet
Switch	10.1.1.5	255.0.0.0	Ethernet

OBSERVATION

The four computers were connected to the network switch using Ethernet cables. Static IP addresses were assigned to all computers within the same network. Connectivity between the computers was verified using the `ping` command. A folder was also shared and accessed from another computer on the same LAN.

CHALLENGES FACED

1. Incorrect IP address configuration may prevent communication between computers.
2. Loose or damaged Ethernet cables may cause connection problems.
3. Firewall settings may block ping requests or file sharing.
4. Incorrect sharing permissions may prevent access to the shared folder.
5. Duplicate IP addresses can cause network communication errors.

RESULT

Thus, a Local Area Network was successfully set up using a network switch and Ethernet cables. IP addresses were assigned to the connected devices, connectivity was verified using the ping command, and file sharing between computers on the LAN was successfully performed.